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003602

August/September 2022

B.Tech. (CE/CSE) VI SEMESTER

Computer Networks (PCC-CS-602)

Time : 3 Hours] [Max. Marks : 75

Instructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART-A

1. (a) Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference? (1.5)

(b) What is the significance of the twisting in twisted-pair cable? (1.5)

(c) A code scheme has a Hamming distance dmin == 4. What is the error detection and correction capability of this scheme? (1.5)

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Question paper

(d) How is the preamble field different from the SFD field? (1.5)

(e) What are the shortcomings of class-full addressing? (1.5)

(f) Which is better DHCP or BOOTP and why? (1.5)

(g) How TCP protocol handles congestion? (1.5)

(h) Name the policies that can prevent congestion. (1.5)

(i) How iterative resolution is different from recursive resolution? (1.5)

(j) How HTTP is same as well as different from SMTP and FTP? (1.5)

PART-B

List out the protocols, data units and communicating devices operating on each layer of OSI Model. (7)

Define spread spectrum and its goal. List any two spread spectrum techniques. (5)

Why is there no need for CSMAlCD on a full-duplex Ethernet LAN? (3)

Find the final code after encoding data bits 1101 into 7-bit even parity Hamming Code? (6)

Why the size of the sender and receiver windows must be at most onehalf of 2 m in selective Repeat ARQ? Explain with an appropriate example. (4)

Compare CSMA/CD with CSMA/CA. (5)

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Question paper

OR

(d) One hundred stations on a pure ALOHA network share a 1-Mbps channel. If frames are 1000 bits long, find the throughput if each station is sending 10 frames per second. (5)

4. (a) An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows: (i) The first group has 64 customers; each needs 256 addresses. (ii) The second group has 128 customers; each needs 128 addresses. (iii) The third group has 128 customers; each needs 64 addresses. Design the sub-blocks and find out how many addresses are still available after these allocations. (10)

(b) List RIP shortcomings and their corresponding fixes. (5)

5. (a) Why transport layer is called an end-to-end layer? Draw a TCP State transition diagram. (5)

(b) What is the value of the receiver window (rwnd) for host A if the receiver, host B, has a buffer size of 5000 bytes and 1000 bytes of received and unprocessed data? Also explain any two techniques to shape traffic? (10)

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6. (a) How IMAP is better than POP3 Protocol? (5)

(b) Explain Request, Response and Entity Headers used in HTTP. (10)

7. Write short note on following :

(i) ALOHA.

(ii) SCTP.

(iii) Firewalls. (15)

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May 2024

B.Tech. (CEDS) (Fourth Semester)

Computer Networks (PCC-CS-602)

Time : 3 Hours] [*Maximum Marks : 75*

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. (a) What is piggybacking ? 1.5

   (b) What are the number of cables required in mesh topology ? 1.5

   (c) The subnet mask for a particular network is 255.255.31.0. Which of the following pairs of IP addresses could belong to this network ? 1.5

   (i) 172.57.88.62 and 172.56.87.233

   (ii) 10.35.28.2 [illegible] 9.4

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(iii) 191.203.31.87 and 191.234.31.88

(iv) 128.8.129.43 and 128.8.161.55

(d) DSSS system spreads the baseband signal by.....the baseband pulses with a pseudo noise sequence. **1.5**

(e) What is the purpose of IEEE 802.11 specifications ? **1.5**

(f) A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal) is 25.6 μs, what is the minimum size of thcce frame ? **1.5**

(g) Station A uses 32 byte packets to transmit messages to Station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use ? **1.5**

(h) What is the purpose of DHCP protocol ? **1.5**

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(i) What is the purpose of SNMP protocol ? 1.5

(j) What is the difference between physical address and logical address ? 1.5

Part B

2. (a) What is the minimum hamming distance of the following strings 010, 011, 101 and 111. 10

(b) Explain the FHSS technique for data transmission. 5

3. (a) Explain the CRC coding and decoding mechanism by taking example of 1001 as dataword in C(7, 4) scheme. 5

(b) Assume in stop and wait ARQ system, the bandwidth of the line is 1 Mbps and 1 bit takes 30 ms to make a round trip. What is the bandwidth delay product ? If the system data frames are 1000 bits in length, what is the utilization percentage of the link. 10

4. (a) Explain why the window size should be < 2 m in Go back n ARQ. where m is the size of the sequence number field in bits. 5

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(b) Draw the flowchart of the CSMA/CD protocol. 5

(c) What is the difference between OSI model and TCP/IP model ? 5

5. (a) What is the difference between UDP and TCP ? 5

(b) A slotted ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces : 10

(i) 1000 frames per second

(ii) 500 frames per second

(iii) 250 frames per second.

6. (a) What is the difference between IPv4 and IPv6 addressing ? 10

(b) Explain the basic concepts of cryptography. 5

7. (a) Explain the token bucket algorithm in detail. 5

(b) Write short notes on the following : 10

(i) DNS

(ii) WWW.

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Sr. No. 011501

December 2023

B.Tech (IT/CSE(AIML)) 5th Semester

Computer Network (PCC-CS-602)

Time: 3 Hours Max. Marks:75

Instructions:
1. *It is compulsory to answer all the questions (1.5 marks each) of Part -A in short.*
2. *Answer any four questions from Part -B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART -A

Q1 (a) Categorize four basic topologies in terms of line configuration. (1.5)

(b) What is the difference between Network layer delivery and transport layer delivery? (1.5)

(c) What is minimum Hamming Distance? Specify minimum Hamming distance for detection of three errors. (1.5)

(d) A network using CSMA/CD has a bandwidth of 10 Mbps. If maximum propagation time is 25.6μs, what is the minimum size of the frame? (1.5)

(e) Why ARP is often cited as Security weakness? (1.5)

(f) A block of address is granted to a small organization. One of the addresses is 205.16.37.39/28. What are the first address, last address and total number of addresses? (1.5)

(g) What is the maximum and minimum size of a TCP Header and why? (1.5)

(h) Name some of the policies that can be used to prevent congestion. (1.5)

(i) What are the functions of two FTP Connections? (1.5)

(j) What are Telnet Options? How they are negotiated? (1.5)

PART -B

Q2 (a) What is TCP/IP Model, explain its layers. Also specify the data units, protocols used and communicating devices used at each layer of TCP/IP Model. (7)

(b) Discuss advantage of Optical Fiber over twisted pair and coaxial cable. (3)

(c) Distinguish between multilevel TDM, multiple slot TDM, and pulse-stuffed TDM (5)

Q3 (a) Explain the working of CSMA/CD protocol. (5)

(b) Compare and contrast Go Back N-ARQ with selective Repeat ARQ. Why the size of sender window must be less than 2^m in Go-Back N-ARQ, where m is the size of the sequence number in bits. (10)

Q4 (a) Compare and Contrast IPV4 and IPv6 packet formats. How can we make transition from IPv4 to IPv6. (5)

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(b) An ISP is granted a block of addresses starting with 150.80.0.0/16. The ISP wants to distribute these blocks to customers as follows. (5)

i. The first group has 256 medium-size businesses; each needs 16 addresses.
ii. The second group has 64 small businesses; each needs 8 addresses.
iii. The third group has 128 households; each needs 4 addresses.

Design the sub-blocks and give the slash notation for each sub-block. Find out how many addresses are still available after these allocations.

(c) List the problems associated with Distance Vector Routing and how they can be resolved? (5)

Q5 (a) Explain the characteristics and frame format of TCP protocol. (7

(b) In a connection, the value of cwnd is 3000 and the value of rwnd is 5000. The host has sent 2000 bytes which has not been acknowledged. How many more bytes can be sent? Also explain the techniques used for traffic shaping. (8

Q6 (a) What is HTTP? Explain its request methods and status codes (5

(b) Explain the protocols used for sending and receiving email (5

(c) Differentiate between Proxy and Packet filter Firewall (5

Q7 Write short note on following (any three) (15

i. Standard Ethernet
ii. Pure Aloha and Slotted Aloha
iii. Connectionless Protocols
iv. Symmetric key cryptography

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